COMMON HALF YEARLY EXAMINATION - 2024 | Reg. No. | | | | | | |

XII - PHYSICS

Time Allowed : 3-00 Hrs. | *Maximum Marks: 70*

Part - I

I. Choose the correct answer: **15 x 1 = 15**

1. An electric field $\vec{E} = 10 \times \hat{i}$ exists in a certain region of space. Then the potential difference $V = V_0 - V_A$, where V_0 is the potential at the origin and V_A is the potential at x = 2m is
   a) 10 V b) –20 V c) +20 V d) –10 V
2. The potential at a point due to a charge of 4×10^{-7} C located at 0.09 m away is
   a) 4×10^4 V b) 4×10^2 V c) 1×10^{-2} V d) 2×10^2 V
3. A wire connected to a power supply of 230 V has power dissipation P_1. Suppose the wire is cut into two equal pieces and connected parallel to the same power supply. In this case power dissipation is P_2. The ratio of P_2 / P_1 is
   a) 1 b) 2 c) 3 d) 4
4. A thin insulated wire forms a plane spiral N = 100 tight turns carrying I = 8 mA (milli ampere). The radii of inside and outside turns are a = 50 mm and b = 100 mm respectively. The magnetic induction at the center of the spiral is
   a) 5 μT b) 7 μT c) 8 μT d) 10 μT
5. A non-conducting charged ring carrying a charge of q, mass m and radius r is rotated about its axis with constant angular speed ω. The ratio of its magnetic moment with angular momentum is
   a) $\frac{q}{m}$ b) $\frac{2q}{m}$ c) $\frac{q}{2m}$ d) $\frac{q}{4m}$
6. A coil of area of cross section 0.5 m^2 with 10 turns is in a plane which is parallel to a uniform magnetic field of 0.2 Wb/m^2. The flux through the coil is
   a) 100 Wb b) 10 Wb c) 1 Wb d) zero
7. In a series resonant RLC circuit the voltage across 100 Ω resistor is 40 V. The resonant frequency ω is 250 rad/s. If the value of C is 4 μF, then the voltage across L is
   a) 600 V b) 4000 V c) 400 V d) 1 V
8. Which of the following is NOT true for electromagnetic waves?
   a) it transports energy
   b) it transports momentum
   c) it transports angular momentum
   d) In vacuum, it travels with different speeds which depend on their frequency
9. For light incident from air on a slab of refractive index 2, the maximum possible angle of refraction is
   a) 30° b) 45° c) 60° d) 90°
10. The transverse nature of light is shown in
    a) interference b) diffraction c) scattering d) polarisation
11. In an electron microscope, the electrons are accelerated by a voltage of 14 kV. If the voltage is changed to 224 Kv, then the de Broglie wavelength associated with the electronic would
    a) increase by 2 times b) decrease by 2 times
    c) decrease by 4 times d) increase by 4 times
12. Nuclear force is
    a) electrostatic force b) long range force
    c) short range force d) magnetic force
13. The ratio between the radius of first three orbits of hydrogen atom is
    a) 1:2:3 b) 2:4:6 c) 1:4:9 d) 1:3:5

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14. To obtain sustained oscillation in an oscillator
    a) feedback should be positive
    b) feedback factor must be unity
    c) phase shift must be 0 or 2π
    d) all the above
15. The technology used for stopping the brain from processing pain is
    a) precision medicine
    b) wireless brain sensor
    c) virtual reality
    d) radiology

Part - II

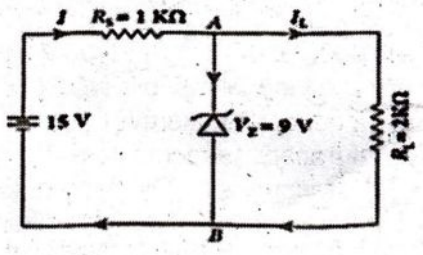
II. Answer any 6 questions. (Q.No.24 is compulsory) **6 x 2 = 12**

16. Write any two applications of capacitors.
17. State Peltier effect.
18. Compute the magnitude of the magnetic field of a long straight wire carrying a current of 1A at distance 1 m from it. Compare it with the Earth's magnetic field.
19. Mention the ways of producing induced emf.
20. What are Fraunhofer lines.
21. Define critical angle.
22. Calculate the distance up to which ray optics is a good approximation for light of wavelength 500 nm falls on an aperture of width 0.5 mm.
23. What is a photo cell? Write its types.
24. Calculate the radius of $_{79}{}^{197}Au$ nucleus.

Part - III

III. Answer any 6 questions. (Q.No.33 is compulsory) **6 x 3 = 18**

25. Derive an expression for the electric potential at a point due to a point charge.
26. Two resistances 4 Ω and 6 Ω are connected in parallel with the 24 V battery. Calculate the equivalent resistance and the values of current in the circuit.
27. Explain the conversion of galvanometer into an ammeter.
28. How will you induce an emf by changing the area enclosed by the coil?
29. Describe Hertz experiment.
30. Two light sources with amplitudes 5 units and 3 units respectively interfere with each other. Calculate the ratio of maximum and minimum intensities.
31. Derive an expression for de Broglie wavelength of electrons.
32. Discuss the alpha decay process with example.
33. Find the current through the Zener diode when the load resistance is 2 kΩ. Use diode approximation.



Part - IV

IV. Answer all the questions. **5 x 5 = 25**

34. a) Deduce the relation for the magnetic field at a point due to an infinitely long straight conductor carrying current using Biot-Savart law. ***(OR)***
    b) Obtain the expression for electric field due to an infinitely long charged wire.
35. a) How the emf of two cells are compared using potentiometer? ***(OR)***
    b) Find the phase relationship between voltage and current in a pure capacitive circuit.
36. a) Explain the types of absorption spectrum. ***(OR)***
    b) Describe the Fizeau's method to determine the speed of light.
37. a) Discuss the interference in thin films and obtain the equationss for constructive and destructive interference for transmitted and reflected light. ***(OR)***
    b) Explain the spectral series of hydrogen atom.
38. a) i) How do we obtain characteristic X-ray spectra?
        ii) Write to the uses of X-rays. ***(OR)***
    b) Draw the circuit diagram of half wave rectifier and explain its working.

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